Answer Key - Machine Learning - Undergraduate Level Assessment 15 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** D
   Options: A) Partitioning based clustering; B) K-means clustering; C) Grid based clustering; D) All of the above
   Note: The correct answer is "All of the above" because spatial clustering algorithms encompass a variety of techniques used to group spatial data points based on their proximity, and all listed options fall within this category.

2. **Answer:** D
   Options: A) True, True; B) False, False; C) True, False; D) False, True
   Note: Overfitting is more likely with a small training dataset because the model can learn noise instead of general patterns, while a small hypothesis space limits the model's complexity and capacity to overfit.

3. **Answer:** C
   Options: A) True, True; B) False, False; C) True, False; D) False, True
   Note: RoBERTa does indeed pretrain on a corpus that is approximately 10 times larger than that used for BERT, making Statement 1 true, while ResNeXT typically employs ReLU activation functions rather than tanh, rendering Statement 2 false.

4. **Answer:** A
   Options: A) linear in K; B) quadratic in K; C) cubic in K; D) exponential in K
   Note: K-fold cross-validation is considered linear in K because the computational cost of training the model increases proportionally with the number of folds, as each fold requires an independent training and validation process.

5. **Answer:** C
   Options: A) Is unbounded, encompassing all real numbers.; B) Is unbounded, encompassing all integers.; C) Is bounded between 0 and 1.; D) Is bounded between -1 and 1.
   Note: The sigmoid function maps any real-valued number into a range between 0 and 1, making the output of a sigmoid node in a neural network always bounded within these limits.

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: The degree of a polynomial in polynomial regression determines the model's complexity, with higher degrees increasing the risk of overfitting by capturing noise in

the training data, while lower degrees may lead to underfitting by failing to capture the underlying trends, thus significantly impacting the trade-off between these two issues.

7. **Answer:** True
   Options: A) True; B) False
   Note: The cost of one gradient descent update is O(D) because the algorithm computes the gradient for each of the D dimensions in the parameter space, requiring a linear amount of computation relative to the number of dimensions.

8. **Answer:** False
   Options: A) True; B) False
   Note: L_2 norm regularization, also known as Ridge regression, shrinks the coefficients towards zero but does not set any coefficients exactly to zero, unlike L_1 norm regularization which can lead to sparse solutions.

9. **Answer:** True
   Options: A) True; B) False
   Note: Bayesians incorporate prior distributions to express beliefs about parameters before observing data, while frequentists rely solely on the data at hand, eschewing prior beliefs in their statistical inference.

10. **Answer:** True
    Options: A) True; B) False
    Note: High entropy signifies a higher degree of disorder or uncertainty in a dataset, indicating that the instances within the partitions belong to multiple classes rather than being predominantly from a single class, thus reflecting a lack of purity in classification.

Long Form Response

11. **Answer:** def subject_marks(subjectmarks): | return subjectmarks
    Note: The provided answer correctly returns the input list of tuples, but it lacks the implementation of sorting using a lambda function, which is essential to fulfill the requirement of the question.

12. **Answer:** def check_literals(text, patterns): | return ('Matched!') | return ('Not Matched!')
    Note: The provided answer correctly identifies the basic structure of a function that uses regex to search for literal strings in a given text, although it lacks the actual implementation details needed to perform the search and return the appropriate match results.

End of Answer Key